

EXHIBIT BUNDLE

METEOROLOGICAL EVIDENCE AND SITE WIND RECORDS

Exhibit references	MO/1 (Met Office certificate); MSL/14 (site anemometer log); LP/11 (lifting plan extract); CR/2 (crane manufacturer's limitation)
Issue	Whether lifting operations should have ceased, when, on whose instruction, and whether wind was causative of the collapse.

PART 1 — MET OFFICE CERTIFICATE (exhibit MO/1)

Certified under s.9 of the Civil Evidence Act 1995 and s.117 of the Criminal Justice Act 2003 as a business record. Station: Leeds Bradford Airport (WMO 03355), 11.8 km NNW of the site, elevation 208 m AOD. Site elevation approx. 34 m AOD.

Time (GMT) 14.11.2023	Mean wind speed	Max gust in period	Direction	Notes
09:00	18 mph	29 mph	245°	-
09:20	19 mph	31 mph	248°	-
10:00	22 mph	34 mph	250°	-
10:20	24 mph	36 mph	252°	-
10:40	25 mph	38 mph	251°	Squall line passage 10:38-10:52
11:00	26 mph	40 mph	254°	-
11:20	27 mph	43 mph	256°	Maximum gust of the day
11:40	24 mph	37 mph	255°	-
12:00	21 mph	33 mph	253°	-

Met Office caveat, reproduced verbatim from the certificate: 'Values are recorded at 10 m above ground at the observing station. Wind speeds at a different location, at a different height above ground, and in a different exposure, may differ materially. Local topography and adjacent structures may increase or decrease both mean speed and gust factor. This certificate should not be treated as a direct measure of conditions at any other location.'

Supplementary opinion of Dr M. Fenwick-Abiodun, consultant meteorologist (instructed by the CPS, report 12.06.2024): allowing for the difference in elevation, the urban roughness of the Hunslet corridor, and the funnelling effect of the river valley, gust speeds at deck level +21.4 m on the east elevation at 11:20 are estimated at between 33 mph and 47 mph, with a central estimate of 39 mph. Dr Fenwick-Abiodun states she cannot narrow the range further and that a reading of 21 mph at that location and height at 11:15 would be 'difficult to reconcile with the regional record, though not impossible if the instrument was sheltered by the building mass or was defective'.

PART 2 — SITE ANEMOMETER LOG (exhibit MSL/14)

Meridian pro forma MSL-F-221. Completed in manuscript. Instrument: handheld vane anemometer, make and serial not recorded. Calibration certificate not produced. Instrument not recovered; Meridian states it was 'lost during site clearance'.

Date	Time	Reading (mph)	Location of reading	Initials	Action taken
14.11.23	07:30	14	Ground level, east compound	RDH	Proceed
14.11.23	09:00	16	Ground level, east compound	RDH	Proceed
14.11.23	10:00	17	Ground level, east compound	RDH	Proceed
14.11.23	11:00	18	Ground level, east compound	RDH	Proceed
14.11.23	11:15	21	Ground level, east compound	RDH	Proceed - below 23 trigger
14.11.23	11:30	19	Ground level, east compound	RDH	-

Observations by HSE: (i) all readings for the day are in a single ink and a single hand, including the 11:30 entry made after the collapse; (ii) no reading is recorded between 07:30 and 09:00 although the pro forma requires hourly readings from commencement of lifting; (iii) the entry at 11:15 is the only one carrying a comment, and the comment refers to the 23 mph trigger; (iv) the pro forma requires readings 'at the working level where practicable'. All readings are recorded at ground level; (v) no entry records the suspension of lifting at 10:40 which Ms Kettleborough says she instructed and which appears in the site daily log (D14).

PART 3 — LIFTING PLAN LP-BW-11 (exhibit LP/11), extract

6.3 "Lifting operations involving cladding panels shall cease when the wind speed measured at the working level exceeds 23 mph (10.3 m/s) gust, or at such lower speed as the Appointed Person considers necessary having regard to the sail area of the

load.”

6.4 “The Appointed Person for this project is the Site Manager, Mr R. Delacroix-Hale.”

6.5 “Where lifting operations are suspended, they shall not recommence except on the express authority of the Appointed Person, recorded in the site daily log.”

6.6 “The panels used on this project (type PC-4, 2.4 m x 1.8 m) present a substantial sail area. The manufacturer of the tower crane (exhibit CR/2, Liebherr 154 EC-H, operating manual s.4.7) specifies a maximum permissible wind speed of 20 mph for loads with a sail area exceeding 4.0 m². The sail area of a PC-4 panel is 4.32 m².”

The conflict: the lift plan trigger (23 mph) is higher than the crane manufacturer's limit for this load (20 mph). The lift plan was approved by Mr Delacroix-Hale as Appointed Person and countersigned on the approval page by Mr Ferris in the box marked 'Director approval — high risk activity'. Mr Ferris's account (D07) is that he had no operational role. He has not been asked in interview about this signature; HSE identified it only on 3 December 2024.

PART 4 — CAUSATION: THE COMPETING POSITIONS

Proposition	Relied on by	Difficulty
Wind was not causative; an untied bay would have failed on any windy day	Dr Vantrease (D19); CPS	Assumes tie absence, which depends on which version of D08 is true
Wind was the sole cause; the bay was adequately tied and an exceptional squall overwhelmed it	Meridian (defence case statement)	No tie found at three anchor positions on survey; no drilled holes
Wind determined the timing of an inevitable failure	Prof. Danecourt (D20.1); HSE structural specialist	Does not answer whether a compliant bay would have survived
Overload was the operative cause and tie absence merely reduced margin	Meridian, in the alternative	Depends on which SWL figure (750 / 1,000 / 2,000 kg) governs